

Video Input

Stage 1
YOLOv8-Seg Detection
(Person Masks + PPE Classes)

Stage 2
MediaPipe
33 Keypoints

Stage 5
Compliance Table | Alert Frames (ZIP) |
Metrics Dashboard

Stage 4
Keypoint-Guided PPE
Association (per worker)

Stage 3
Skeleton-Statistics
Tracker

Body Part Attention Regions:
Helmet → Head (nose, eyes, ears)
Vest → Torso (shoulders, hips)
Gloves → Wrists | Boots → Ankles

Skeleton Signature Vector:
① Normalised joint positions
② Bone length ratios
③ Joint angles
④ Mean $\Delta(x,y)$ velocity over K frames
⑤ $\text{cost} = \alpha \cdot \text{IoU} + \beta \cdot \|s_i - s_j\|$